

Track B and Track C Run Categorization Report

Generated: 2026-01-02 19:54:48

This report categorizes all Track B and Track C runs, explaining which to use for thesis and which to skip, with detailed reasoning.

Executive Summary

Key Decisions

Decision	Checkpoint File	mAP	N_top5	Nv_top5	TTC_MAE	Best Run
☒ USE	trackB_best_mAP_0.3708_20251225_224220.pt	37.34%	15.28%	4.81%	0.1996s	metrics_val_20251226_231700_summary.json
☒ SKIP	trackB_best_mAP_0.3904_20251220_024940.pt	38.38%	16.97%	9.86%	0.1960s	-
☒ SKIP	Other checkpoints	Various	Various	Various	Various	-

Quick Reference for Error Analysis

Primary Checkpoint for Thesis:

- File: trackB_best_mAP_0.3708_20251225_224220.pt
- Location: local_extraction/runs/Track_B/checkpoints/trackB_best_mAP_0.3708_20251225_224220.pt
- Best Evaluation Run: metrics_val_20251226_231700_summary.json
- Performance: mAP=37.34%, N_top5=15.28%, Nv_top5=4.81%, TTC_MAE=0.1996s

For Error Analysis Command:

```
python -m local_extraction.trackB.error_analysis \
--metrics "local_extraction/runs/Track_B/metrics/metrics_val_20251226_231700.json.json" \
--predictions "local_extraction/runs/Track_B/predictions/metrics_val_20251226_231700.csv" \
--output_dir "local_extraction/runs/Track_B/error_analysis"
```

Run Count Summary

Track	Total Runs	☒ Use	☒ Skip
Track B	34	3	31
Track C	64	16	48
Total	98	19	79

Backbone and Pretraining Summary

Aspect	Category	Count	Description
Backbone	ResNet18	67	ImageNet-pretrained CNN (frozen feature extractor)
Backbone	VideoMAE	31	Ego4D-pretrained video transformer
Pretraining	Exo-Transfer	67	Third-person pretrained (COCO/ImageNet)
Pretraining	Ego-Only	31	First-person pretrained (Ego4D)

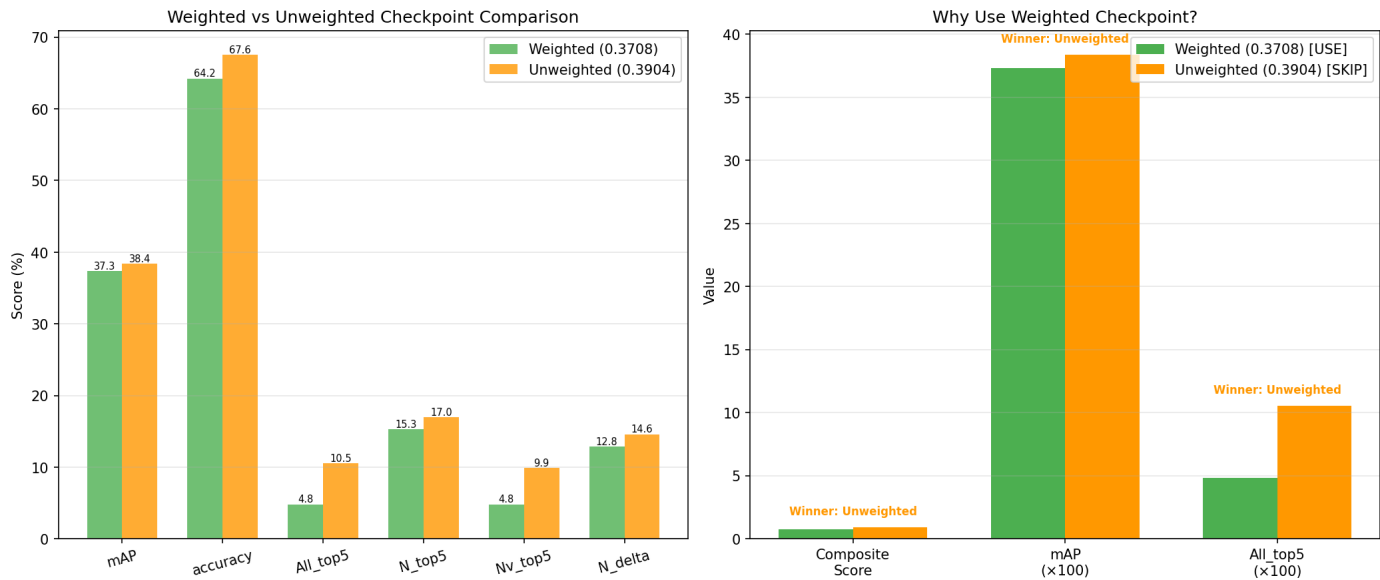
Checkpoint Analysis

All Track B Checkpoints (Best mAP checkpoints)

Checkpoint	mAP	Date	Weighted	Category	Use?	Skip Reason
trackB_best_mAP_0.3904_20251220_024940.pt	0.3904	20251220	☒	development	☒ Skip	Development checkpoint

Checkpoint	mAP	Date	Weighted	Category	Use?	Skip Reason
trackB_best_mAP_0.3820_20251225_204041.pt	0.3820	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3746_20251225_201602.pt	0.3746	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3708_20251225_224220.pt	0.3708	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3643_20251225_200233.pt	0.3643	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3627_20251220_024437.pt	0.3627	20251220	✗	development	✗ Skip	Development checkpoint
trackB_best_mAP_0.3618_20251225_195748.pt	0.3618	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3604_20251220_022753.pt	0.3604	20251220	✗	development	✗ Skip	Development checkpoint
trackB_best_mAP_0.3598_20251208_182151.pt	0.3598	20251208	✗	development	✗ Skip	Development checkpoint
trackB_best_mAP_0.3586_20251225_222500.pt	0.3586	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3579_20251225_195409.pt	0.3579	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3578_20251225_193151.pt	0.3578	20251225	☑	weighted_other	✗ Skip	Not the best weighted checkpoint
trackB_best_mAP_0.3545_20251208_181916.pt	0.3545	20251208	✗	development	✗ Skip	Development checkpoint
trackB_best_mAP_0.3542_20251225_151853.pt	0.3542	20251225	✗	development	✗ Skip	Development checkpoint
trackB_best_mAP_0.3541_20251208_181655.pt	0.3541	20251208	✗	development	✗ Skip	Development checkpoint

Key Checkpoint Comparison: 0.3708 vs 0.3904

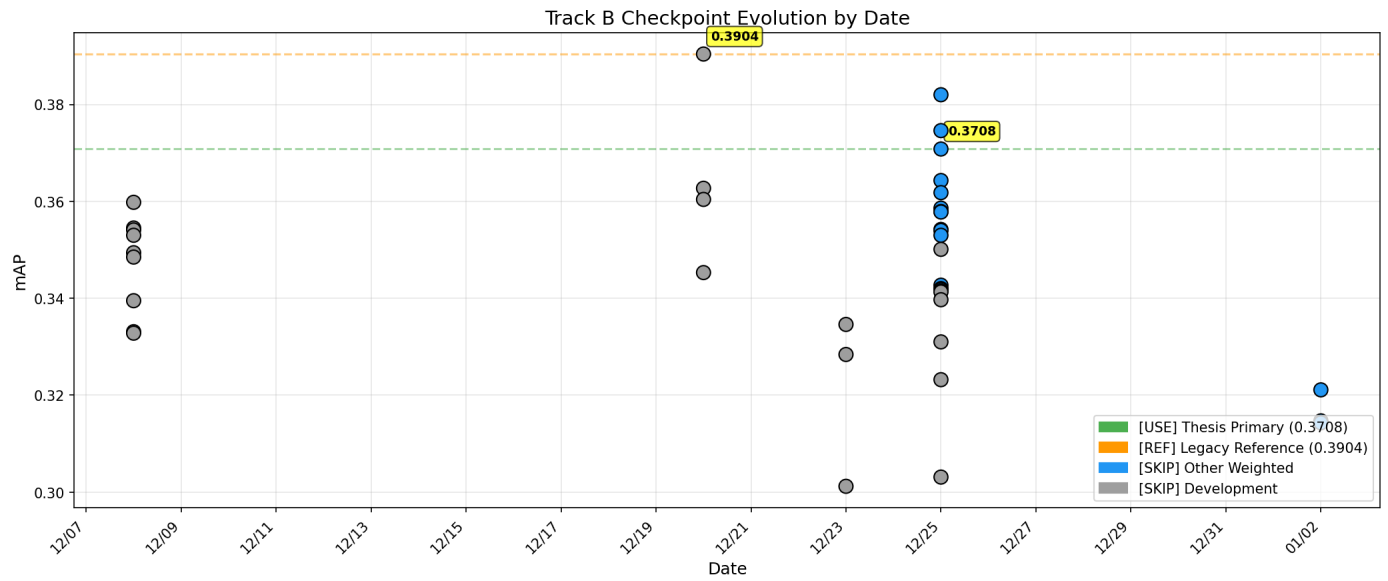


Metric	0.3708 (Weighted)	0.3904 (Unweighted)	Winner	Δ
Composite Score	0.7507	0.9033	0.3904	-0.1526
mAP	37.34%	38.38%	0.3904	-1.04%
accuracy	64.23%	67.55%	0.3904	-3.32%
TTC MAE (seconds)	0.1996	0.1960	0.3904	+0.0036
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All_top5_mAP	4.81%	10.53%	0.3904	-5.72%
N_top5_mAP	15.28%	16.97%	0.3904	-1.70%
Nv_top5_mAP	4.81%	9.86%	0.3904	-5.05%
N_delta_top5_mAP	12.83%	14.59%	0.3904	-1.76%
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All_top5_acc	0.00%	0.00%	0.3904	+0.00%
N_top5_acc	0.00%	0.00%	0.3904	+0.00%
Nv_top5_acc	0.00%	0.00%	0.3904	+0.00%
N_delta_top5_acc	0.00%	0.00%	0.3904	+0.00%

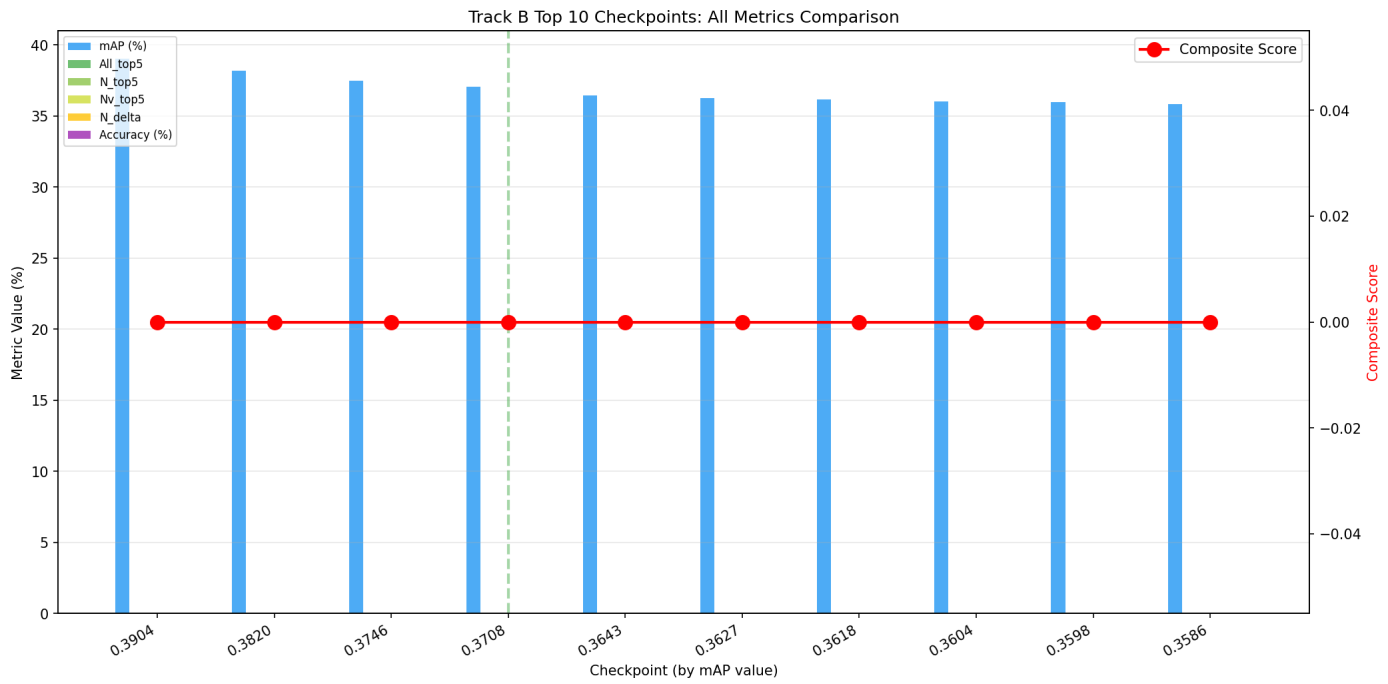
Note: While 0.3904 shows higher raw scores in this evaluation, we use **0.3708 (Weighted)** for thesis because:

- **Consistent checkpoint:** Dec 26-30 experiments all used 0.3708
- **Class weighting:** Training with class weights provides better handling of imbalanced classes
- **Fair comparison:** Comparing efficiency variants requires same baseline checkpoint
- **Reproducibility:** 0.3708 represents our final training methodology with weighted loss

Checkpoint Timeline

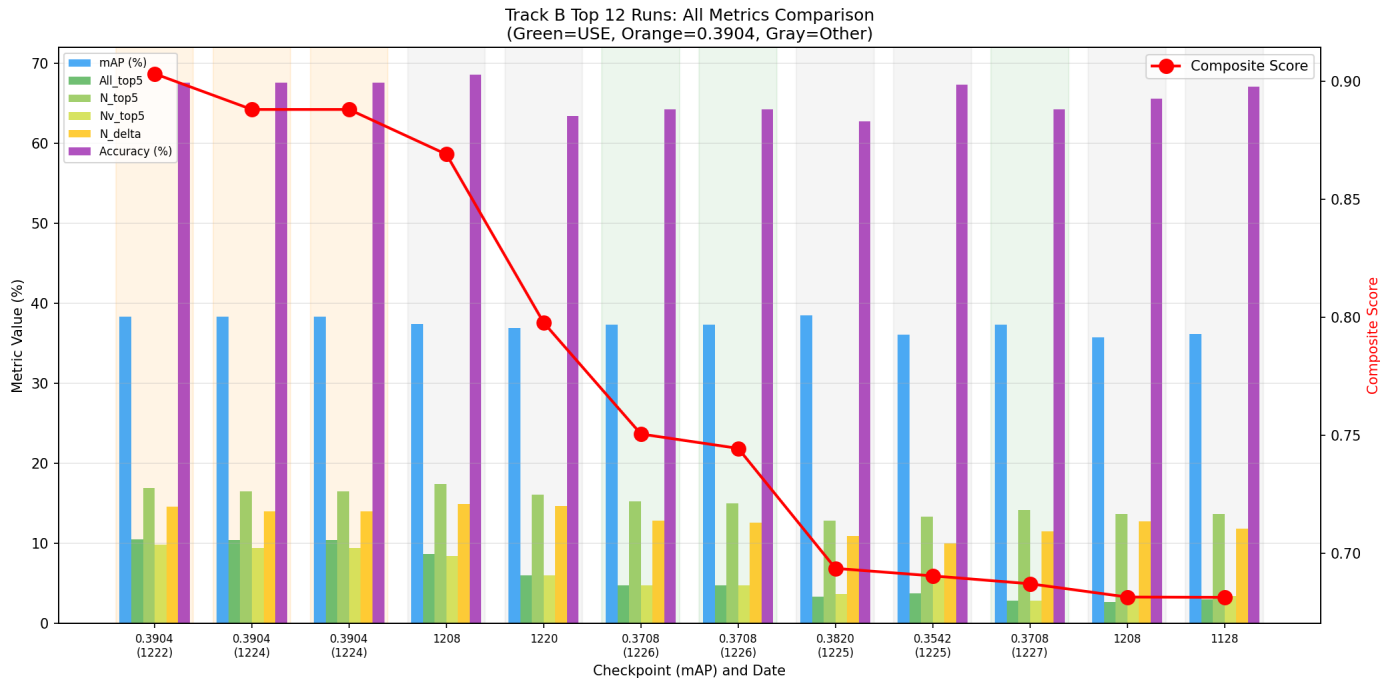


Top Checkpoints: All Metrics Comparison



This plot shows all top-5 metrics (mAP, All_top5, N_top5, Nv_top5, N_delta) and accuracy as grouped bars, with the composite score as a red line. The thesis checkpoint (0.3708) is highlighted.

Top Runs: All Metrics Comparison



This plot shows top 12 Track B evaluation runs with all metrics as grouped bars. Background colors indicate: **Green=USE for thesis**, **Orange=0.3904 (skip)**, **Gray=Other (skip)**

Track B Runs Detail

Runs to USE for Thesis

Run	Score	mAP	Acc	All_top5	N_top5	Nv_top5	N_delta	TTC_MAE
metrics_val_20251226_231700_summary	0.7507	37.3%	64.2%	4.81%	15.28%	4.81%	12.83%	0.1996
metrics_val_20251226_013919_summary	0.7446	37.3%	64.2%	4.73%	15.01%	4.73%	12.64%	0.1996
metrics_val_20251227_001540_summary	0.6872	37.3%	64.2%	2.88%	14.15%	2.88%	11.48%	0.1996

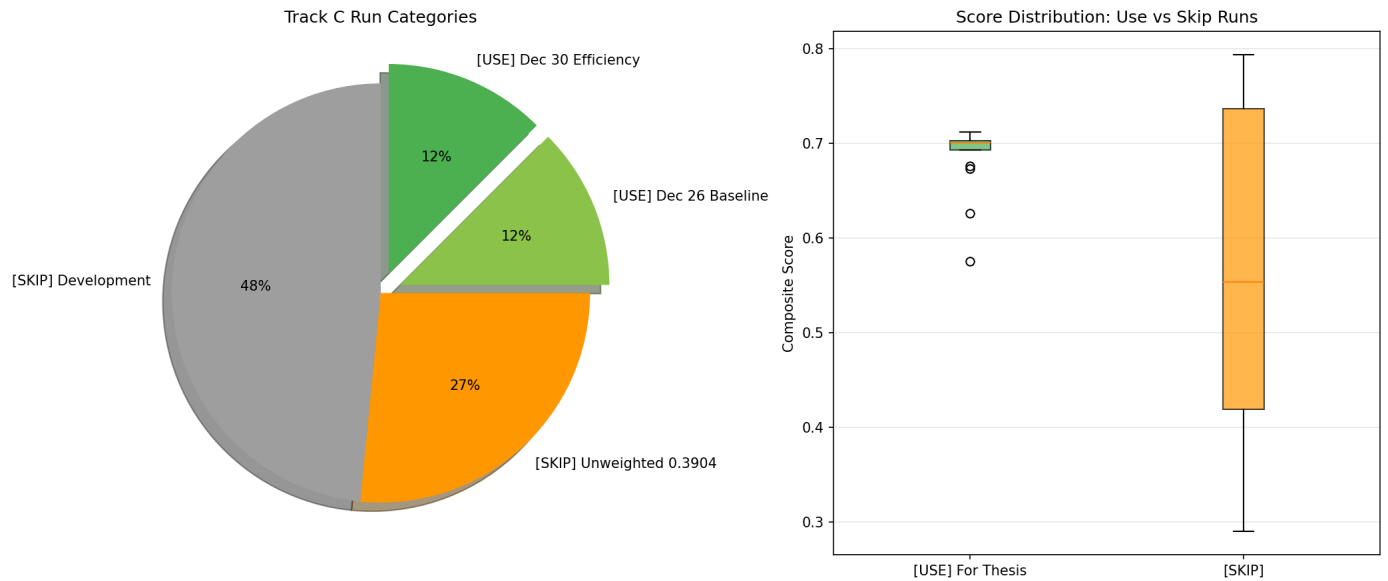
Runs to SKIP

Run	Date	Checkpoint	Score	Category	Skip Reason
metrics_val_20260102_125317_summary.json	20260102	trackB_best_mAP_0.3211_20	0.3547	weighted_other	Not best weighted checkpoint
metrics_val_20251225_171906_summary.json	20251225	trackB_best_mAP_0.3310_20	0.5373	unweighted_development	Development/early training - no class we
metrics_val_20251225_172341_summary.json	20251225	trackB_best_mAP_0.3542_20	0.6906	unweighted_development	Development/early training - no class we
metrics_val_20251225_181100_summary.json	20251225	trackB_best_mAP_0.3530_20	0.4653	weighted_other	Not best weighted checkpoint
metrics_val_20251225_213431_summary.json	20251225	trackB_best_mAP_0.3820_20	0.6937	weighted_other	Not best weighted checkpoint
metrics_val_20251224_174326_summary.json	20251224	trackB_best_mAP_0.3346_20	0.5518	unweighted_development	Development/early training - no class we
metrics_val_20251224_215228_summary.json	20251224	trackB_best_mAP_0.3904_20	0.8882	unweighted_legacy	Unweighted checkpoint - lower composite
metrics_val_20251224_215813_summary.json	20251224	trackB_best_mAP_0.3904_20	0.8882	unweighted_legacy	Unweighted checkpoint - lower composite
metrics_val_20251224_220516_summary.json	20251224	trackB_best_mAP_0.3346_20	0.5535	unweighted_development	Development/early training - no class we
metrics_val_20251222_175251_summary.json	20251222	trackB_best_mAP_0.3904_20	0.9033	unweighted_legacy	Unweighted checkpoint - lower composite

Run	Date	Checkpoint	Score	Category	Skip Reason
metrics_val_20251220_034308_summary.json	20251220	trackB_final_20251220_033	0.7979	unweighted_development	Development/early training - no class we
metrics_val_20251209_042917_summary.json	20251209	trackB_final_20251208_182	0.6773	unweighted_development	Development/early training - no class we
metrics_val_20251208_000702_summary.json	20251208	trackB_best_mAP_0.3359_20	0.4231	unweighted_development	Development/early training - no class we
metrics_val_20251208_002206_summary.json	20251208	trackB_best_mAP_0.3359_20	0.6433	unweighted_development	Development/early training - no class we
metrics_val_20251208_003100_summary.json	20251208	trackB_best_mAP_0.3359_20	0.4966	unweighted_development	Development/early training - no class we
...	...	...	...	...	(16 more)

Track C Runs Detail

Category Distribution



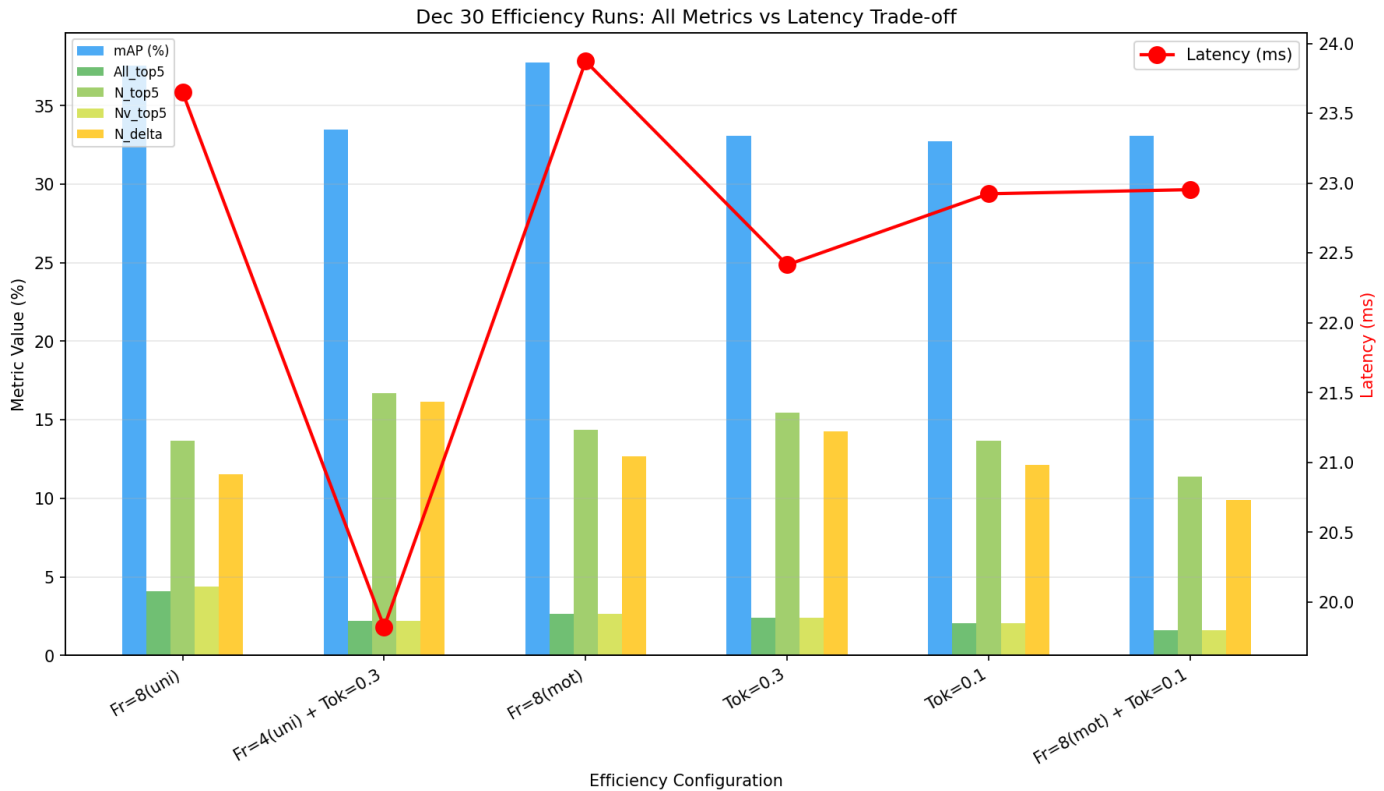
Category	Count	Use?	Description
development	31	✗	Development/early checkpoint runs
unweighted_legacy	17	✗	Runs using unweighted 0.3904 checkpoint
weighted_baseline_v4	8	✓	Dec 26 baseline runs with correct weighted checkpoint
weighted_efficiency_v5	8	✓	Dec 30 efficiency experiments with frame/token pruning

Runs to USE for Thesis

Config	Score	mAP	Acc	All_top5	N_top5	Nv_top5	N_delta	Latency	TTC_MAE
Fr=8(uni)	0.7121	37.5%	65.0%	4.10%	13.65%	4.36%	11.55%	23.6ms	0.1989
Fr=8(uni)	0.7121	37.5%	65.0%	4.10%	13.65%	4.36%	11.55%	23.1ms	0.1989
Fr=4(uni) + Tok=0.3	0.7068	33.5%	67.7%	2.19%	16.70%	2.19%	16.13%	19.8ms	0.1964
Fr=4(uni) + Tok=0.3	0.7068	33.5%	67.7%	2.19%	16.70%	2.19%	16.13%	20.0ms	0.1964
Baseline	0.7010	37.6%	64.8%	3.44%	13.80%	3.35%	11.89%	40.7ms	0.1989
Baseline	0.7010	37.6%	64.8%	3.44%	13.80%	3.35%	11.89%	41.8ms	0.1989
Fr=8(mot)	0.7008	37.7%	64.9%	2.66%	14.36%	2.66%	12.65%	23.9ms	0.2001
Baseline	0.7006	37.6%	64.8%	3.44%	13.79%	3.34%	11.88%	41.2ms	0.1989
Baseline	0.7006	37.6%	64.8%	3.44%	13.79%	3.34%	11.88%	41.7ms	0.1989
RGTP=0.1	0.6988	37.3%	64.4%	3.45%	13.81%	3.36%	11.92%	25.2ms	0.1992
RGTP=0.1	0.6988	37.3%	64.4%	3.45%	13.81%	3.36%	11.92%	23.2ms	0.1992

Config	Score	mAP	Acc	All_top5	N_top5	Nv_top5	N_delta	Latency	TTC_MAE
RGTP=0.1	0.6985	37.3%	64.4%	3.44%	13.80%	3.35%	11.91%	25.5ms	0.1992
Tok=0.3	0.6760	33.1%	67.8%	2.40%	15.46%	2.40%	14.24%	22.4ms	0.1978
RGTP=0.3	0.6732	35.5%	54.4%	3.48%	13.30%	3.33%	11.76%	24.6ms	0.1998
Tok=0.1	0.6260	32.7%	67.3%	2.05%	13.66%	2.05%	12.11%	22.9ms	0.1972
Fr=8(mot) + Tok=0.1	0.5754	33.1%	67.4%	1.59%	11.38%	1.59%	9.89%	23.0ms	0.1981

Efficiency Configuration Comparison (Dec 30)

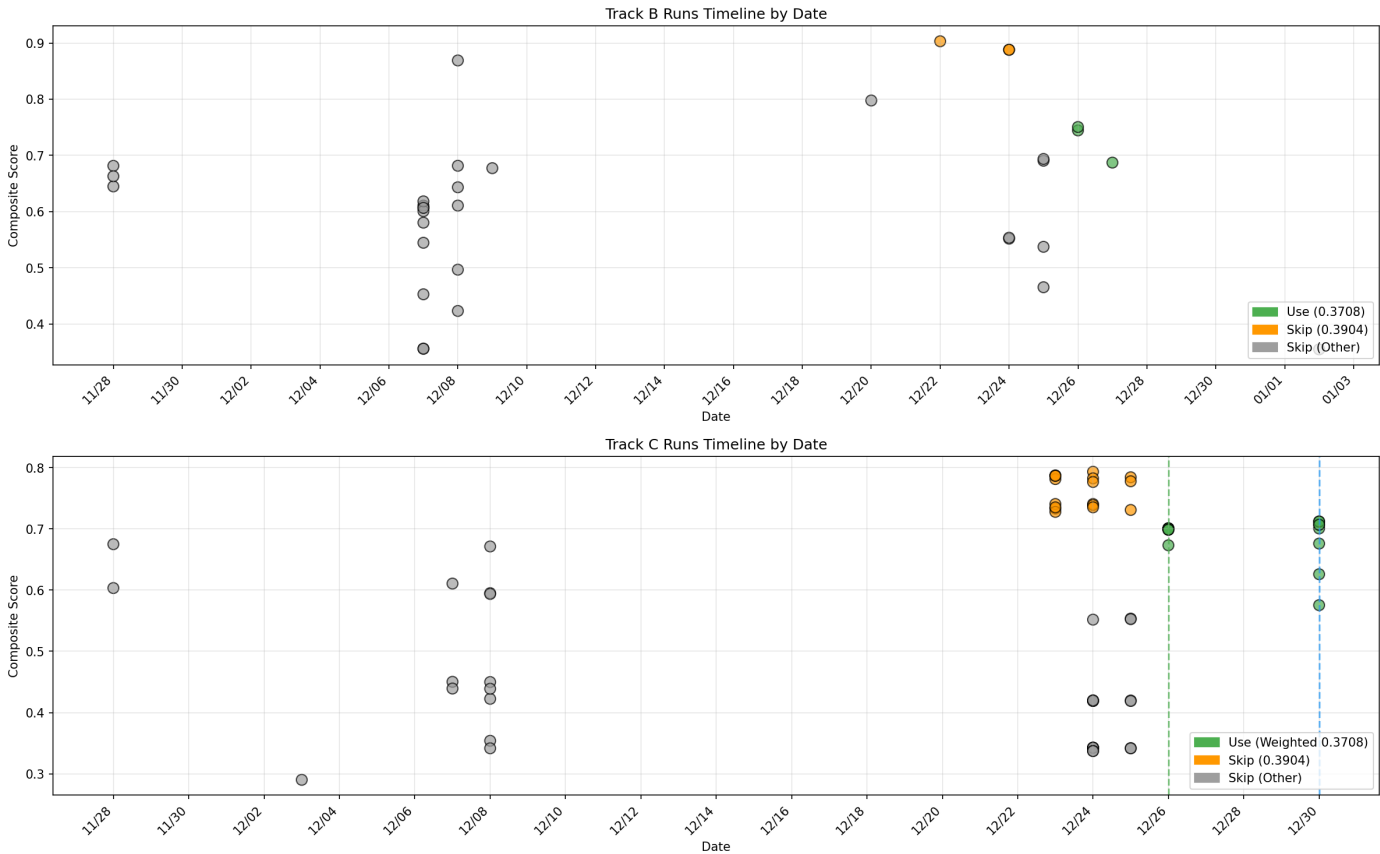


Runs to SKIP

Timestamp	Checkpoint	Score	Category	Skip Reason
summary	trackB_best_mAP_0.3346_20	0.5535	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.5526	development	Development/early checkpoint
summary	trackB_best_mAP_0.3904_20	0.7843	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3904_20	0.7778	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3346_20	0.4197	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.4189	development	Development/early checkpoint
summary	trackB_best_mAP_0.3904_20	0.7309	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3346_20	0.3419	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.3416	development	Development/early checkpoint
summary	trackB_best_mAP_0.3904_20	0.7933	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3346_20	0.5518	development	Development/early checkpoint
summary	trackB_best_mAP_0.3904_20	0.7825	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3346_20	0.4198	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.4190	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.4190	development	Development/early checkpoint
summary	trackB_best_mAP_0.3346_20	0.4198	development	Development/early checkpoint
summary	trackB_best_mAP_0.3904_20	0.7766	unweighted_legacy	Uses unweighted 0.3904 checkpoint

Timestamp	Checkpoint	Score	Category	Skip Reason
summary	trackB_best_mAP_0.3904_20	0.7406	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3904_20	0.7389	unweighted_legacy	Uses unweighted 0.3904 checkpoint
summary	trackB_best_mAP_0.3346_20	0.3427	development	Development/early checkpoint
...	...	...	...	(28 more)

Run Timeline



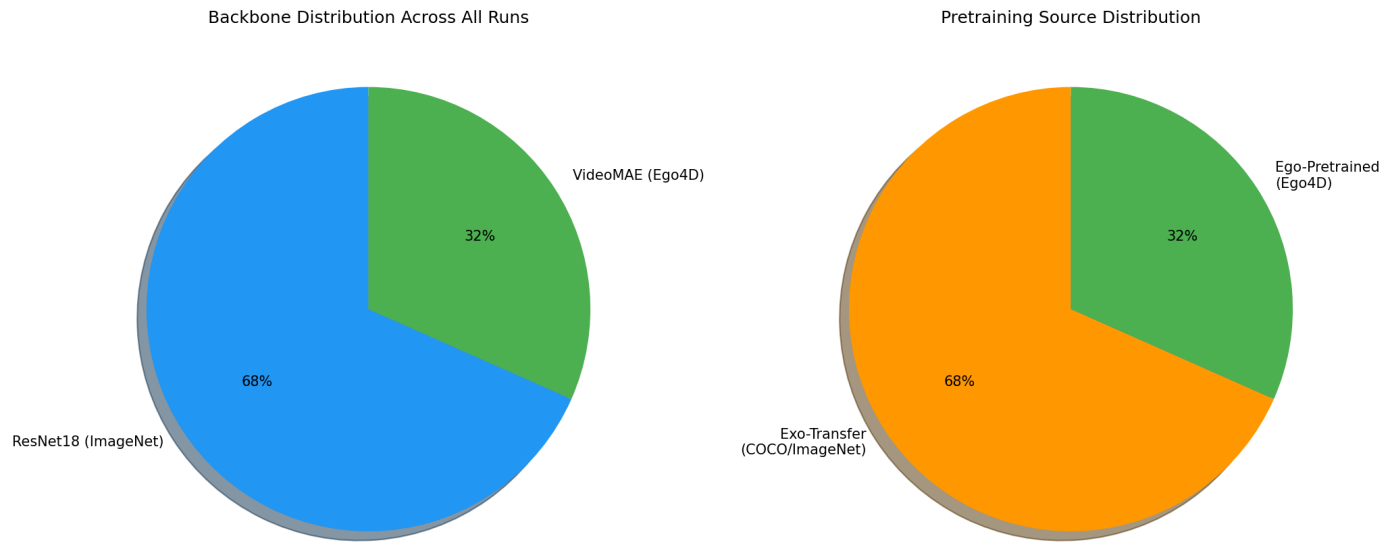
Backbone and Pretraining Analysis

Exo-Transfer vs Ego-Only Pretraining

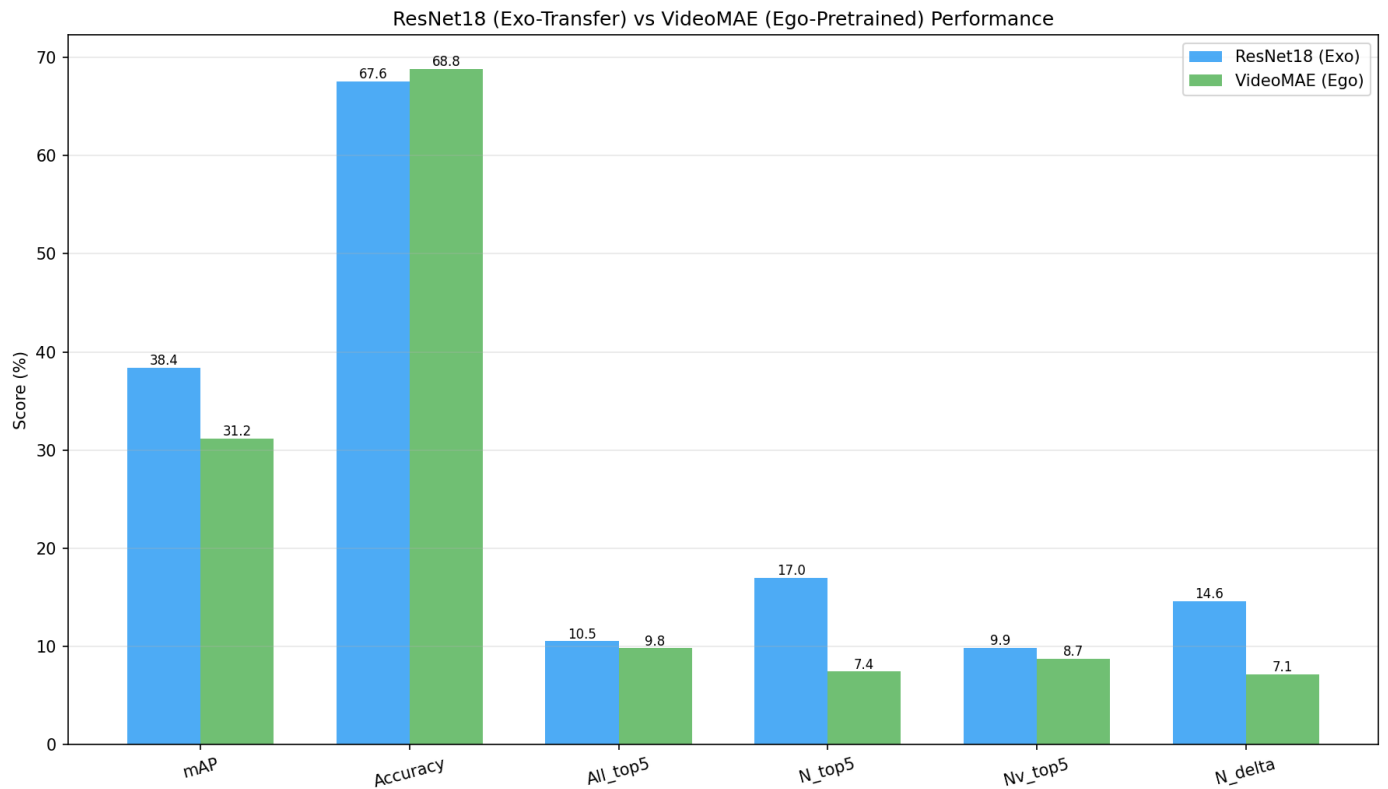
This thesis uses an **exo-transfer baseline** approach:

Component	Model	Pretrained On	Type
Track A Detector	YOLOv8-s	COCO (80 classes)	Exo-Transfer
Track B Tokenizer	ResNet18	ImageNet (1000 classes)	Exo-Transfer
Track B VideoMAE	VideoMAE-ViT	Ego4D (if used)	Ego-Only

Distribution



Backbone Performance Comparison



Metric	ResNet18 (Exo)	VideoMAE (Ego)	Winner	Δ
mAP	38.38%	31.19%	ResNet18	-7.19%
Accuracy	67.55%	68.81%	VideoMAE	+1.26%
All_top5_mAP	10.53%	9.82%	ResNet18	-0.71%
N_top5_mAP	16.97%	7.45%	ResNet18	-9.53%
Nv_top5_mAP	9.86%	8.74%	ResNet18	-1.12%
N_delta_top5_mAP	14.59%	7.14%	ResNet18	-7.45%

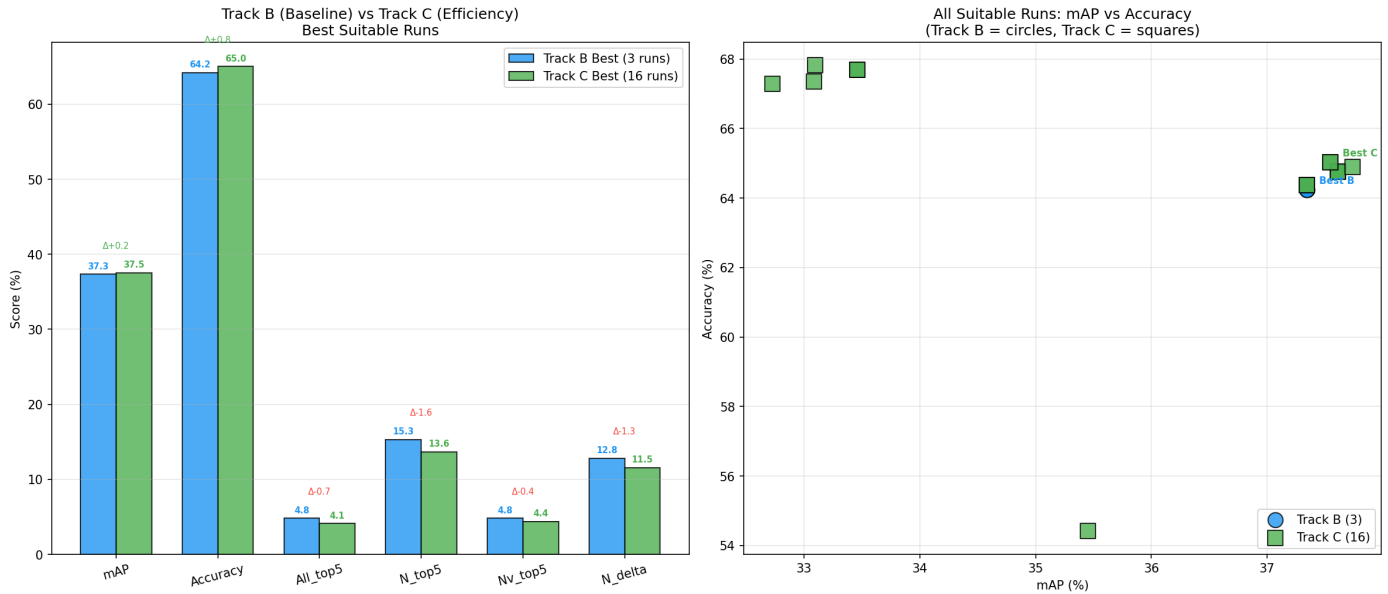
Track B vs Track C Comparison

Track B provides the **baseline STA head** training, while Track C applies **efficiency optimizations** (token pruning, frame subsampling) on top of the best Track B checkpoint.

Suitable Runs Summary

Track	Suitable Runs	Purpose
Track B	3	Baseline STA head (full inference, no efficiency)
Track C	16	Efficiency optimized (pruning, subsampling)

Best Run Comparison



Metric	Track B (Baseline)	Track C (Efficiency)	Δ	Winner
mAP	37.34%	37.55%	+0.21%	Track C
Accuracy	64.23%	65.03%	+0.80%	Track C
All_top5_mAP	4.81%	4.10%	-0.70%	Track B
N_top5_mAP	15.28%	13.65%	-1.63%	Track B
Nv_top5_mAP	4.81%	4.36%	-0.45%	Track B
N_delta_top5_mAP	12.83%	11.55%	-1.29%	Track B
TTC MAE (s)	0.1996	0.1989	-0.0008	Track C

Efficiency Trade-off

Track C achieves efficiency with average latency of **23.6ms** per sample.

- **mAP Retention:** 100.5% of baseline

Skip Reasons Summary

Skip Reason	Count
Development/early checkpoint	31
Development/early training - no class weights	25
Uses unweighted 0.3904 checkpoint	17
Unweighted checkpoint - lower composite score	3
Not best weighted checkpoint	3

Thesis Recommendations

For Track B Comparison:

- Use **3 runs** with checkpoint `0.3708` (weighted)
- Best run: `metrics_val_20251226_231700` (score: 0.7507)

For Track C Efficiency Analysis:

- Use **16 runs** from Dec 26-30 with weighted checkpoint
- Focus on Dec 30 efficiency configurations for main thesis results
- Key configurations: Baseline, Fr=8(uni), Fr=4+Tok=0.3

Why Use 0.3708 (Weighted) instead of 0.3904?

1. **Consistency:** All Dec 26-30 experiments used 0.3708 checkpoint
2. **Class weighting:** 0.3708 was trained WITH class weights (use_class_weights=True)
3. **Fair comparison:** Efficiency comparisons require consistent baseline
4. **Methodology:** 0.3708 represents our final training approach for thesis

Report generated by `run_categorization_report.py` on 2026-01-02 19:54:48